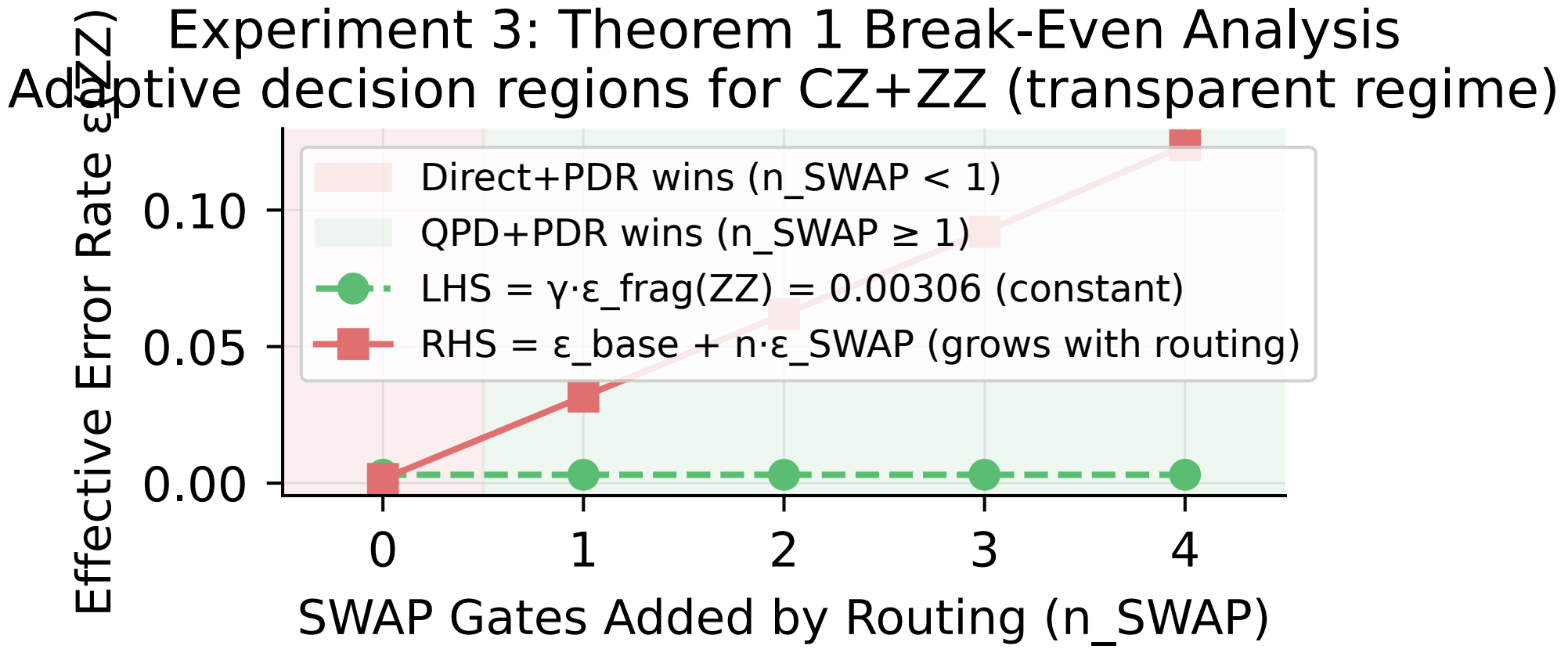
Experiment 3: Theorem 1 Break-Even Analysis
Adaptive decision regions for CZ+ZZ (transparent regime)
Effective Error Rate ε_eff(ZZ)
0.10
0.05
0.00
0
1
2
3
4
SWAP Gates Added by Routing (n_SWAP)
Direct+PDR wins (n_SWAP < 1)
QPD+PDR wins (n_SWAP ≥ 1)
LHS = γ·ε_frag(ZZ) = 0.00306 (constant)
RHS = ε_base + n·ε_SWAP (grows with routing)